Committee: International Telecommunication Union (ITU)

Topic: The Question of Ensuring the Sustainable Management of Earth's Orbit by Mitigating Space Debris and Optimizing Frequency Allocation

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Introduction

Today, with sophisticated aerospace technology, outer space is considered the second home for humanity. Just like Earth, space is a shared place for all member states. Not only rocket engineering, but also launching satellites has revolutionized global telecommunications, connecting humanity closer than ever before.

When humanity first approached outer space, space debris was not considered seriously. In the vast universe, space debris was just a negligible material. However, while the whole universe might be infinite, the space that humanity can use is exclusively limited. Space debris does not simply disappear into distant deep space. Instead, it is trapped by the Earth's gravitational pull and remains in specific orbits, rotating around the Earth. This debris continues to circle around Earth's orbit, becoming a new danger that threatens space missions and operations. While the waste on Earth is governed by each member state's national regulations within each nation's boundaries, space debris lacks post-processing systems after use. Despite its potential dangers, there are no binding international frameworks that clarify the responsibility for managing space debris.

These unmanaged physical threats directly affect the global infrastructure, particularly in telecommunications. The increasing volume of space debris interferes with observation and navigation of Earth, making interaction between Earth and outer space harder. Debris with high velocity can impact operating satellites, paralyzing telecommunication systems. Furthermore, the accumulation of space debris accelerates the advent of the Kessler Syndrome, which is a tipping point where chain reactions of collisions cover the whole orbit of Earth, making it unusable for decades or even centuries

Ensuring a sustainable Earth orbit involves confronting challenges beyond physical threats like space debris. Indeed, the systemic structure of frequency allocation managed by the ITU has also posed another challenge. ITU adheres to the “first-come, first-served service”, which allocates frequencies primarily to the nations that register first. These countries are mostly the MEDC countries that have launched rockets for a long time. Initially, this policy was implemented to incentivise innovation, maximise efficiency, and prioritise the countries that are ready for immediate launch. However, as the world has entered the new space age, the competition among countries, and even private enterprises, has intensified.

As these physical and regulatory challenges continue to increase, multilateral cooperation and continuing interest among the international community are more crucial than ever in order to ensure the sustainable management of the Earth's orbit.

Definition of Key Terms

Space Debris

Space debris refers to all human-driven waste in outer space orbits. Space debris is artificial masses that are not currently in operation and will not be in the future. This includes everything from defunct satellites and fragments of broken spacecraft to tools dropped by astronauts during spacewalks.

LEO Satellite Communication

LEO satellite communication refers to a network infrastructure that transmits data using artificial satellites positioned at LEO (low Earth orbit), which is a relatively low altitude of 200 to 2,000km from the Earth's surface. Operating much closer to Earth than traditional GEO (Geostationary Earth Orbit) satellites, which orbit at approximately 36,000km, LEO systems provide significant advantages. They allow low communication latency. While LEO satellites can bridge the digital divide for developing nations, they can also create unnecessary signal overlap or frequency interference with existing terrestrial networks.

ASAT (Anti-satellite Weapon)

ASAT refers to a specialized space-warfare weapon designed to destroy or neutralize an operational satellite, thereby paralyzing its communication capabilities. These weapons are operated by both kinetic destruction and non-kinetic defeat, such as jamming and cyberhacking.

Spectrum Warehousing

Spectrum warehousing is to preemptively lock down valuable frequencies and orbital slots, such as those in LEO, in competition with global competitors to attract investments or get the property rights for future profit. This practice artificializes resource scarcity and prevents latercomer nations and legitimate operators of space debris from accessing vital frequencies. It accelerates competition, friction between countries, and ultimately inequality among member states.

Paper Satellites

Paper satellites are satellites that exist only on official ITU paperwork but are never practically manufactured, launched, or used. Filing paper satellites is one way of Spectrum warehousing. They are usually filed to block competitors, preempt rights, or even to resell frequencies at a premium.

Radio Frequency Interference (RFI)

Radio frequency interference, also known as RFI, can be conceptualized as similar to distinct radio stations overlapping. The generation of harmful interference is predominantly due to unwanted emissions, regulatory infringements, and intentional interference. Unwanted emissions of frequency occur as inevitable physical byproducts during transmission. Furthermore, operational misconduct and infringements of the radio regulations, such as unauthorized emissions and operation of non-coordinated frequency assignments, or even deliberate jamming, are also a significant driving force.

The Kessler Syndrome

Formulated in 1978 by scientist Donald J. Kessler, the Kessler Syndrome explains that once the volume and the density of space debris in LEO surpass a threshold, artificial objects will collide in a chain. In this state, collisions will generate more debris, which leads to more frequent collisions. This loop, in turn, generates an uncontrollable amount of space debris, leaving future generations with the work to clean it.

High Seas Principle

The High Seas Principle was first used in UNCLOS Part VII - High Seas, an international agreement that clarifies that international waters belong to all nations equally and cannot be owned exclusively by any single country. It clarifies that every member state has the freedom of navigation, overflight, and fishing in international waters. In this chair report, the High Seas Principle is applied not only to waters but also to outer space orbits. As applied to water, space orbits are not subject to ownership, and all states have the right to access them freely.

The Outer Space Treaty

In this chair report, the Outer Space Treaty was considered by the legal subcommittee in 1966, and agreement was reached in the general assembly in the same year. The Outer Space Treaty provides the basic framework for international space law. Namely, the Outer Space Treaty includes principles such as

- the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries and shall be the province of all mankind;
- outer space shall be free for exploration and use by all States;
- outer space is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means;
- States shall not place nuclear weapons or other weapons of mass destruction in orbit or on celestial bodies or station them in outer space in any other manner;
- the Moon and other celestial bodies shall be used exclusively for peaceful purposes;
- astronauts shall be regarded as the envoys of mankind;
- States shall be responsible for national space activities whether carried out by governmental or non-governmental entities;
- States shall be liable for damage caused by their space objects; and
- States shall avoid harmful contamination of space and celestial bodies.

Mega Constellation

A mega constellation is a group of many satellites that work together for the purpose of delivering broadband internet access. Mega constellations can comprise hundreds or even thousands of satellites. It is mostly established in LEO rather than the traditional high orbits.

Active Debris Removal

Active Debris Removal is a process of directly removing space debris with physical technologies, such as sending a spacecraft to rendezvous with, capture, and deorbit orbital waste. It refers to direct physical cleanup of orbital waste to mitigate collision risks in LEO. Unlike passive compliance measures, like the 25-year rule, ADR addresses objects already in orbit that pose a collision risk.

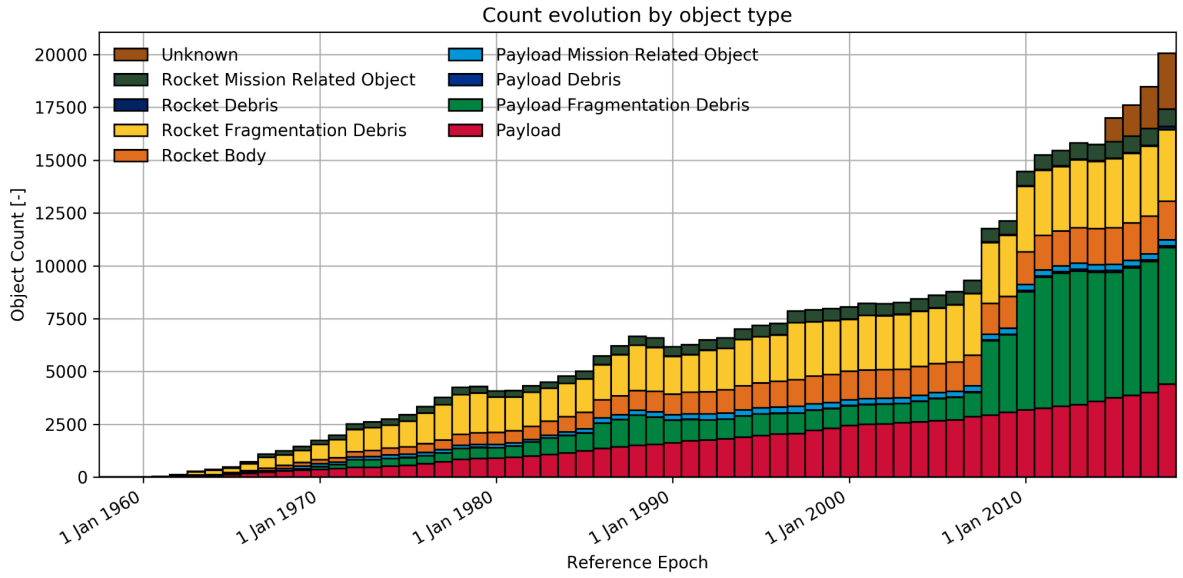
CBMs

Confidence Building Measures (CBMs) are actions that reflect goodwill toward exchanging information with an adversary. The purpose of such measures is to decrease misunderstanding, tension, fear, and conflict between parties by emphasizing trust. They can be outlined in four main types: communication, constraint, transparency, and verification. For instance, entities may establish hotlines or require prenotification and information exchange. Space technologies, such as active debris removal, are inherently dual-use. To prevent militarization and ensure peaceful usage of outer space, building confidence and trust between nations is essential. Protocols like pre-launch notifications, sharing of orbital tracking data, and informing others before practicing ADR technologies can build mutual trust and prevent frictions.

Background Information

Humanity has utilized outer space for more than 60 years of active space missions. Satellites and other spacecraft infrastructure remain vital to modern civilization, developing space science, supporting Earth observations, telecommunications, navigation, and human space exploration. The immense volume of space objects, including both operating satellites and space debris, possesses a mass exceeding 17,400 tonnes. The majority of space debris is heavily concentrated within LEO. LEO, also known as low Earth orbit, is now becoming the world's largest garbage dump. There are currently millions of pieces of space debris flying in LEO. Most of this space debris is composed of fragmented pieces of spacecraft, parts of spent rockets, abandoned satellites that are no longer working, and leftover particles from explosions. The threat from these objects is due to their extreme velocity. Most space debris orbits at 18,000 miles per hour on average, which is a velocity even faster than a bullet. At this kinetic energy level, even a centimetre-sized fragment can completely obliterate an active satellite.

Space debris has been derived exclusively from over 60 years of human space activities. About 24 percent of catalogued objects are satellites, including both operational and defunct ones. In addition, mission-related objects, such as lens covers and launch adapters, account for 11 percent of space debris. While only 7 events have been directly associated with collisions, the majority of space debris has been generated by in-orbit explosions of spacecraft and rocket bodies. Leftover fuel or batteries in discarded rocket stages and dead satellites heat up over time, leading to explosions.



The causes of space debris by object type- The European Space Agency

ASAT tests and satellite collisions are the following causes of space debris. High-speed impacts generate massive amounts of debris. Most notably, there was the kinetic destruction of the Chinese Fengyun-1C spacecraft during an anti-satellite missile test in 2007, and the accidental collision between the American Iridium 33 and the defunct Russian Cosmos 2251 spacecraft in 2009. The two incidents generated thousands of fragments, instantly increasing the large orbital debris population in LEO by up to 70 percent and increasing the risks of daily collision.

As such fragments slowly accumulate in orbit, they push the environment toward a catastrophic collapse, the Kessler syndrome.

Hard kills of ASAT contaminate orbit for a period of time and can trigger chain reactions of surrounding satellites(Kessler Syndrome). Despite international warnings from bodies like the G7 and EU, major states justify their tests under national security, highlighting the lack of international regulations on counter-space weapons. Consequently, addressing space debris is no longer only a scientific but also a geopolitical challenge.

The Limitation of Active Debris Removal

Despite the nature of this crisis, there are currently no integrated international space laws that mandate or enforce cleaning up existing space debris. Implementing a remedy remains a significant financial and technological challenge. Removing space junk is extraordinarily expensive, especially when removing the 6,000 tons of uncontrolled material in the LEO region. Ultimately, space debris is not the responsibility of any single nation. It is the shared, collective responsibility of every country on Earth. Managing and mitigating space debris is not only a profound international challenge, but also an opportunity for establishing collaborative frameworks for preserving the outer space environment.

As such, mitigating current and future generations of space debris alone is insufficient to stabilize LEO and prevent the Kessler syndrome, and active removal is necessary. However, implementing Active Debris Removal (ADR) raises profound legal and political paradoxes. Without a modernized framework, technical cleanup will be impossible.

First, there is a fundamental legal ambiguity regarding the obligation to remove debris under current international space law. Previously established frameworks such as the UN COPUOS Space Debris Mitigation Guidelines and IADC guidelines are non-binding soft law. While these guidelines recommend post-mission disposal, they lack enforcement systems and mandatory regulations. Furthermore, ADR is prevented by strict jurisdiction under the Outer Space Treaty, Article VIII. Article VIII claims that a launching state retains absolute jurisdiction and ownership over its registered space objects. As a result, no third parties can legally regulate or remove these objects without obtaining prior consent from the original state. One-sided removal of another nation's space debris, especially defunct satellites, is a direct violation of national sovereignty.

The ADR technology itself is also a barrier. The 1972 Liability Convention complicates operational authorization. Under Article VI of the OST, states have international responsibility for all national space activities, including those conducted by non-governmental commercial entities. If an ADR encounters technical failure, the removing state can be held liable for the damage. For example, if failures cause secondary fragmentation or damage an operating satellite, the state faces Fault-based liability. This high legal risk prevents states and commercial entities from undertaking ADR missions. In addition, the dual usage of ADR technologies creates severe space security problems. Technologies required for orbital cleanup are functionally identical to ASAT weapons. Because these systems can also be used to disable and destroy operational foreign military assets, spacefaring nations view ADR deployments with suspicion. Without further improvement of international treaties, ADR could be misinterpreted as offensive space weaponization.

Paper Satellites and Spectrum Warehousing

The radio frequency spectrum and satellite orbits are fundamentally limited natural resources. Because these resources are not infinite, the ITU highlights that all member states must possess equal and equitable rights to utilize them. The primary goal of the ITU is to unite global telecommunications so that radio communication services can operate without disruptions to one another.

To manage the finite resources of orbital and radio frequencies, the ITU has historically relied on a first-come, first-served administrative system. Under the mechanism, rights are granted only to states that file their satellite plans first. However, paradoxically, this has led to over-competition for monopolizing low-Earth orbit and frequencies. Preempting LEO (low-Earth orbit), the prime location for operating satellites, is important for every member state in order to maintain technical strength and

national security. As a result, paper satellites have proliferated. By filing massive amounts of satellites, each member state secures LEO earlier than others. One strategy of filing paper satellites is splitting filings. For instance, when mega constellation operators like SpaceX's Starlink file for licenses for orbit, they avoid registering their infrastructure as a single network. Instead, they split the application into dozens of smaller filings in order to bypass the power threshold restrictions, bulk up their satellite count, and preempt the frequency. In November 2021, the Rwanda government submitted an ITU filing to use 337,320 satellites under the project "Cinnamon-937." This single application demanded more satellites than the total number that currently exist in Earth's orbit that currently exist in Earth's orbit. The true goal behind this massive filing was Spectrum warehousing.

In response to the growing criticism of the first-come, first-served system, the ITU has made structural attempts to prevent such administrative abuses and speculative filings. ITU holds the World Radio Communication Conference (WRC) every 3-4 years, and updates the Radio Regulations through discussions. Through revising the Radio Regulations, ITU copes with the most current arguments, including paper satellites and space debris. Most notably, WRC-19 in 2019 adopted a new innovative milestone-based regulatory approach for the deployment of non-geostationary satellite orbits. The milestone framework is to gradually set goals for a long-term project. Every nation's filings are required to deploy 10 percent of their constellations within 2 years after the end of the bringing-into-use period, 50 percent within 5 years, and complete deployment within 7 years. This will provide a regulatory mechanism to help ensure that ITU's frequency allocation reflects the actual deployment of satellites, and ultimately prevent the mass production of paper satellites that are abandoned for the long term.

Despite these regulatory interventions by the ITU, structural limitations still exist. The Milestone framework can fundamentally worsen competition among nations to preempt orbital and frequency resources rather than promote equitable distribution. Furthermore, administrative loopholes still allow spectrum warehousing, especially by private enterprises.

As a consequence, the original frequency allocation system of the ITU has proven to be increasingly ineffective. All member states should seek solutions that not only prevent such administrative side effects of ITU regulations, but also pursue fundamental innovation that addresses the root cause of spectrum warehousing, such as the first-come, first-served system, and ensure sustainable, equitable frequency allocation.

Regulatory Gap of Private Enterprises

In the past, space exploitation was done exclusively by nations, but the New Space Era shifted the driving force to private companies such as SpaceX, OneWeb, and Amazon. Private operators now launch thousands of satellites annually. However, this commercial boom presents unprecedented environmental, scientific, and legal challenges due to severe regulatory gaps.

Current space governance relies on outdated international frameworks that are merely high seas law, and objects are governed solely by their launching state. While the Outer Space Treaty mandates that states must authorize and continue to supervise non-governmental space activities, the treaty did not anticipate the advent of mega constellations. Consequently, national compliance with space debris mitigation remains significantly low. Instead of proactively requesting debris prevention, governments often just sign agreements to cover any financial liability if an accident happens. As a result, companies continue operating space objects with low risk and fewer restrictions.

Furthermore, the geopolitical effect of private enterprises has created an issue. Under general international law such as the ARSIWA, private actions are not directly attributable to the state. However, because the Outer Space Treaty strictly assigns state responsibility for private space ventures, military deployment of commercial satellites is not clearly addressed, blurring the line between commercial actions and state involvement.

As such, the House highlights the implementation of new international treaties and laws that specify the regulations and responsibilities not only applied to states but also to private companies.

Possible solutions

Enhancing Governance and Space Traffic Coordination

To address the rapid growth of objects in orbit, international institutions and national governments must establish clearer governance and coordination systems. Effective space governance requires strengthening the role of international bodies, such as the ITU and UNOOSA. Reforming administrative processes of such institutions is required to prevent spectrum warehousing and to ensure transparent communication among launching authorities. Furthermore, international space traffic coordination will allow nations to share real-time tracking data, which prevents physical collisions and RFI in LEO.

Strengthening National Regulatory Standards and Post-mission Disposal

Member states are encouraged to adopt non-binding international guidelines into enforceable domestic regulations to ensure effective compliance with space debris disposal systems. National Space Agencies can also mandate that all licensed satellite operators adopt post-mission disposal and concentrate on developing debris removal technologies.

Fostering CBMs for Active Debris Removal

Active Debris removal technologies require robust international cooperation and mutual trust. A primary challenge of ADR is the nations' mistrust due to its dual-use as ASAT counterspace weapons. To decrease security concerns and prevent geopolitical conflict, states must establish Confidence-Building Measures. Under frameworks supported by the UN and the ITU, launching states can mandate transparency protocols, namely pre-mission notifications, real-time sharing of orbital tracking data, and continuous observation of operating ADR. By establishing CBMs in international space governance, member states can build trust, ensuring active debris removal initiatives are executed as cooperative environmental technology, rather than a threat to national security. In addition, member states can establish multilateral ADR operations, which will ease conducting high-risk ADR missions under the UN's authorization rather than solely shifting the burden of responsibility to the member states.

Prevention of Spectrum Warehousing and Paper Satellites

To remove the administrative overhead of allocating frequency spectrum and ensure equitable access for every member state, reforming the fundamental allocation system of the ITU- the first-come, first-served method is essential. This may include methods such as requiring stricter standards and information when filing, escalating financial deposits and filing fees when paper satellites are caught, and strengthening international supervision among countries launching mega constellations. However, since such methods escalate launching risks and regulatory barriers to nations and private entities, blindly restricting the regulations of frequency allocation might lead to discouragement and slowdown of the aerospace industry, making ADR technologies also hard to develop. Delegates must formulate a revised framework of frequency allocation while ensuring the original objectives of the first-come, first-served system, which are fostering space industrialization and preventing RFI.

Regulatory Gaps in Private Enterprises and Emerging Nations

To prevent private mega constellation operators from abusing the regulatory gaps of outer space usage and waste liability, each member state may require private entities to obtain mandatory orbital deposits and invest a particular amount of their revenue in post-mission disposal and debris removal systems, and emphasise the Polluter pays principle. Furthermore, to ensure these standards do not also burden emerging spacefaring nations and space debris removal startups, MEDCs can facilitate capacity-building programs and transfers of active space debris removal technologies.

Major parties involved

USA

The USA stands as the structural core of the modern global space industry and the pioneer of commercial space technology. Within federal entities like NASA and dominant private enterprises, the USA aggressively pursues space superiority as its primary national objective. US firms manufacture 83% of all commercial satellites launched globally in 2025. As the International Space Station decommissions by 2030, the U.S. seeks to entirely replace it with commercially operated space stations owned by American private companies.

The U.S. space policy is concentrated on accelerating domestic markets, noting that its first preference is for commercial solutions. Recently, the U.S. Federal Communications Commission (FCC) shortened the 25-year period for decommissioned satellite disposal to 5 years. While this administrative decision actively supports the safe and environmentally friendly use of satellites, it also may serve as a potent regulatory weapon for control over global private space enterprises. Since foreign operators seeking access to the U.S. market must follow FCC rules, global actors are forced to comply with the domestic law of the U.S., raising the concern of U.S. domestic law superseding international space law. Furthermore, the U.S. dominates in the technologies within its National Aeronautics and Space Administration (NASA). Being the pioneer of modeling space debris and establishing the standards of fragment extraction and reorbiting technologies, NASA is one of the most significant scientific agencies that initially recognized the Kessler syndrome. In addition, the US Space Force (USSF) provides the structural data for tracking space debris to other nations.

China

In contrast to the commercialized paradigm of the United States, China executes a highly centralized and state-led doctrine. They aim to secure military and national security in outer space. For instance, China possesses a distinct operational space station called the Tiangong space station. Recently, China actively deployed state-sponsored mega-constellation initiatives such as Guowang and the G60 Starlink, which are designed to secure critical orbital tracks such as LEO. However, China raised international worries due to its high contribution to orbital debris. In LEO, Chinese space debris accounts for 4,234 out of 10,723 tracked objects, accounting for 40 percent of the total. The most significant generation of Space debris was the Chinese anti-satellite missile test conducted in

2007. This intentionally destroyed its weather satellite, Fengyun-1C, and generated over 3,400 pieces of debris. While the Inter-Agency Space Debris Coordination Committee(IADC) and the major spacefaring nations adhere to the 25-year orbit guideline, China's implementation of international orbital guidelines remains limited. As China plans to rapidly accelerate its launches to deploy mega constellations, it can escalate the threat to global orbital safety.

Russia

As a traditional space dominator, the Russian Federation officially aligns with UN space debris mitigation guidelines. However, Russia's actions based on national security and defense have created threats to the LEO environment. For instance, in 2021, its direct-ascent ASAT missile test created over 1,500 trackable debris fragments. Russia claims that the debris made by missile tests quickly dissipates. It advocates for non-binding and voluntary guidelines, but rejects Western-led space regulations that might prevent its space strategic capabilities.

ESA

The European Space Agency, consisting of France, the United Kingdom, and 21 other countries, is a major international pioneer of space safety. ESA aims to eliminate the generation of new space debris by 2030 through its "Zero Debris Approach" based on the Zero Debris Charter. ESA is an organization dedicated to space exploration and orbital sustainability. Despite addressing active remediation in such ways, the agency confronts technical and financial problems in removing its own debris from orbit. These limitations are due to the voluntary regional frameworks, which lack legally enforceable mandates and guaranteed long-term funding mechanisms, especially for costly ADR.

Rwanda

As a prominent representative of emerging spacefaring nations, Rwanda highlights the growing tension between orbital sustainability and equitable access for developing nations. While acknowledging the hazards posed by space debris, Rwanda and other developing nations argue that strict, costly debris mitigation measures should not become market barriers that block their legitimate space ambitions. They advocate for flexible international frameworks that guarantee equitable spectrum allocation, ensuring that latecomer nations retain the right to utilize space for national economic development.

Republic of Korea

The Republic of Korea represents a rapidly growing spacefaring nation, utilizing a hybrid strategy of public and private models. With the establishment of the Korea AeroSpace Administration(KASA) and major technical milestones, such as the domestic launch of vehicle Nuri(KSLV-II), the lunar orbiter Danuri, and the Korean Positioning System(KPS) project, South Korea seeks to expand commercial and technological presence in outer space. Alongside emerging space powers like the United Arab Emirates(UAE), the Republic of Korea supports reforming the ITU's traditional first-come, first-served system in frequency allocation. On the other hand, the Republic of Korea emphasizes that frequency regulations must remain flexible and accessible, ensuring that strict administrative thresholds do not create barriers for latecomer nations and commercial startups.

India

India stands as a major spacefaring nation among developing nations, operating under the Indian Space Research Organization(ISRO). Possessing independent launch capabilities and commercial satellites, India seeks to balance aggressive commercial expansion and LEO's sustainability. India advocates for reforming the ITU's traditional first-come, first-served frequency allocation system and argues that MEDCs must not monopolize orbits, at the expense of latecomer nations.

Private companies

The rapid commercialization of outer space has made private entities vital in both creating and mitigating space debris. Mega constellation operators such as SpaceX create a significant amount of space debris in Low Earth Orbit. On the other side, specialised commercial ADR startups, such as ClearSpace and Astroscale, are pioneers of technical solutions that utilize robotic arms and magnetic capture to clear hazardous space debris. However, these companies still have burdens and hurdles due to responsibility risks and third-party property rights.

Timeline Of Events

Date	Description of Event
The Outer Space Treaty Agreement, 1967	The Outer Space Treaty Agreement was discussed in 1966 and issued in 1967. The agreement was reached in the General Assembly on the basic framework for international space law.
Formulation of the Kessler Syndrome, 1978	Scientist Donald J. Kessler formulates the theory of exponential collision chain reactions of space debris in LEO.
Chinese Anti-Satellite Missile Test, 2007	China destroys its weather satellite, Fengyun-1C, generating thousands of pieces of space debris.
Iridium-Cosmos Satellite Collision, 2009	The American Iridium 33 collides with the defunct Russian Cosmos 2251, significantly escalating orbital debris risks.
Rwanda's "Cinnamon-937" ITU Filing, November 2021	Rwanda submits an ITU filing for 337,320 satellites, highlighting the issue of spectrum warehousing.
Russian ASAT Missile Test, 2021	Russia conducts a direct-ascent ASAT missile test, creating over 1,500 trackable debris fragments.
FCC Shortens Disposal Timeline, 2022	The U.S. Federal Communications Commission reduces the satellite disposal timeline from 25 years to 5 years.

UN Involvement, Resolutions, Treaties and Events

UN Involvement

The United Nations' involvement in ensuring the sustainable management of Earth's orbit has been active, primarily driven by policy-making and specialized bodies such as the General Assembly (UNGA), the Committee on the Peaceful Uses of Outer Space (UNCOPUOS), and the International Telecommunication Union (ITU). The Legal Subcommittee of UNCOPUOS has served as a central forum for negotiating foundational legal principles and international treaties governing outer space. Additionally, agencies like the ITU manage the finite global common resources of radio frequency spectrum and satellite orbital slots. Furthermore, the ITU regularly holds conferences, the World Radiocommunication Conferences (WRC), revising and reviewing the Radio Regulations. The next WRC (WRC-27) will be held in 2027 in Shanghai, China.

UN resolutions

- Resolution 2222 (XXI), 19 December 1966, clarifies the principles of governing the activities of states in the exploration and use of outer space.

- Resolution 2777 (XXVI), 29 November 1971, the Convention on International Liability for Damage Caused by Space Objects states that countries that launch the spacecraft definitely are liable for damage caused by their space objects.

- Resolution 62/217, 22 December 2007, the International cooperation in the peaceful uses of outer space endorses the Space Debris Mitigation Guidelines of COPUOS and ensures the sustainable, equitable use of outer space.

UN Treaties

- **Space Liability Convention** provides that a launching state shall be absolutely liable to pay compensation for damage caused by its space objects on the surface of the Earth or to aircraft, and liable for damage due to its faults in space. The convention also provides for procedures for the settlement of claims for damages.

- **The Outer Space Treaty, December 19, 1966 (Resolution 2222 (XXI))**

Considered by the Legal Subcommittee in 1966 with an agreement reached in the General Assembly in the same year, the Outer Space Treaty was concluded. It provides the fundamental framework for international space law. The treaty establishes that space exploration is for the benefit of all mankind, outer space is free from national appropriation, the Moon is reserved for peaceful purposes, and astronauts are envoys of humanity. Crucially, it prohibits placing nuclear weapons or weapons of mass destruction in orbit, establishes that states are strictly liable for damage caused by their space objects, assigns international responsibility for both governmental and non-governmental national activities, and mandates that states must avoid the harmful contamination of outer space and celestial bodies.

UN Instruments

- **IADC Space Debris Mitigation Guidelines** were established initially in 2002, october. This includes the 25-year post-mission satellite disposal guideline. This guideline is established with the purpose of promoting efforts to deal with space debris issues, in such ways as preventing explosive and collisional on-orbit break-up, and limiting the objects released during normal operations.

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